

Hybrid Retrieval Design

Hybrid RAG combines semantic retrieval and keyword retrieval to improve evidence selection.

Semantic retrieval uses embeddings to find chunks that are related in meaning, even when exact words do not match.

BM25 keyword retrieval uses exact term matching and inverse document frequency to capture names, tools, technical terms, and codes.

Reciprocal Rank Fusion merges both result lists without requiring scores to share the same scale.

Reranking and Confidence

After RRF, candidates should be reranked before answer generation.

A lightweight local reranker can score query term coverage, semantic rank, keyword rank, and whether both retrieval methods agree.

Retrieval confidence can combine normalized RRF score, reranker score, query term coverage, and source diversity.

High confidence means the document evidence is strong enough for grounded answering.

Grounded Answer Rule

When web search is disabled, the assistant must answer only from uploaded or seeded documents.

If the documents do not contain enough evidence, the assistant should say it cannot answer from the available PDFs.

This rule makes the system more predictable, testable, and suitable for demos where source control matters.